



Introduction to Blockchain

CSE-355

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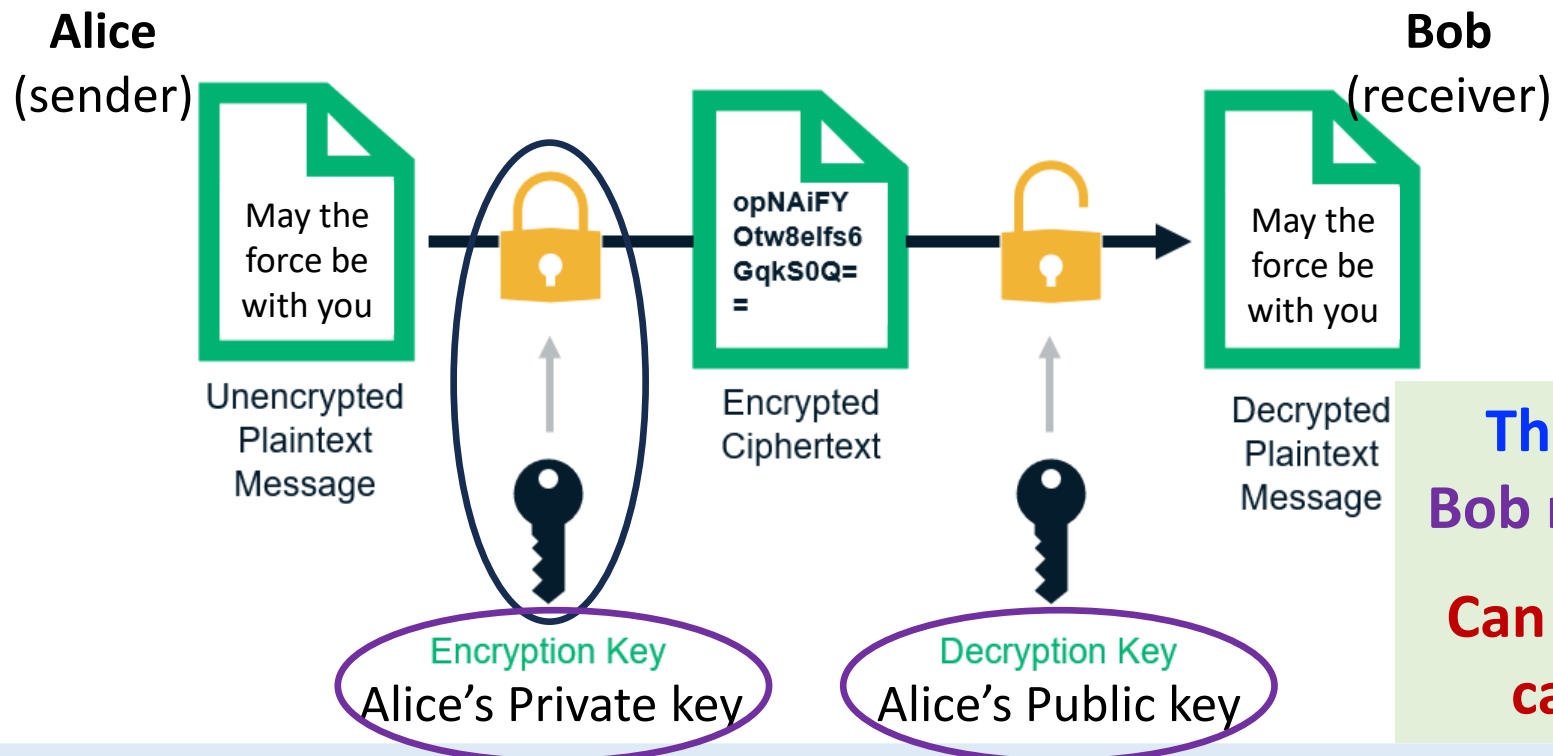
Spring 2026

Lecture 05

Tuesday, Feb 03, 2026

Public Key Cryptography

Each user generates a **public-private key pair**
Public keys: **very public** Private keys: **very private**



Thought Question:
Bob receives some data

Can he now be sure it came from Alice?

- Alice (encrypts) or **locks** the transaction
- 1) Using her **private key** –
In **addition**, Alice can **ALSO** OPTIONALLY (why?)
 - 2) Use Bob's public key – Only Bob's private key can **unlock**

Digital Signatures: A simple-minded approach

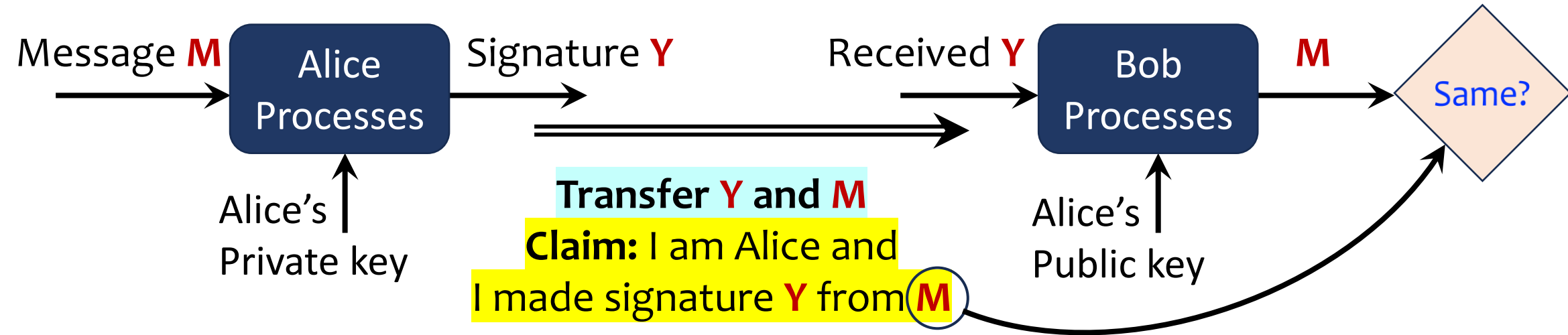
- Alice “signs” with her own private key
 - Signing is the same process as encrypting/encoding
- Alice sends to Bob – a signed document
 - The original document
 - The signatures
- Bob uses Alice’s public key to decode the signature and compares
- If comparison works, this [probably?] verifies:
 - It was Alice who sent the message (authentication/non-repudiation)
 - The message was not forged (integrity)
- Anyone else can also decode the message
 - And that may also be ok
 - What if we don’t want anyone else to decode the message?

PKI and Transactions on Bitcoin Blockchain

- Cryptocurrency owned by Alice
 - From a previous transaction [by someone who paid her]
 - Value locked [previously] to Alice's PUBLIC key
- Alice unlocks the entire value in the transaction
 - Using her PRIVATE key
 - No one else can do it
- Alice makes the transaction
 - Locks some value to Bob's PUBLIC key (pays to public key!)
 - Only Bob will be able to unlock using his PRIVATE key
 - **Important:** Alice locks the remaining value to her own PUBLIC key
 - **Important:** Alice can't deny making the transaction later on. Why?

Digital Signatures: Integrity + Authentication

GOAL: Check for Message integrity and Sender's Authenticity

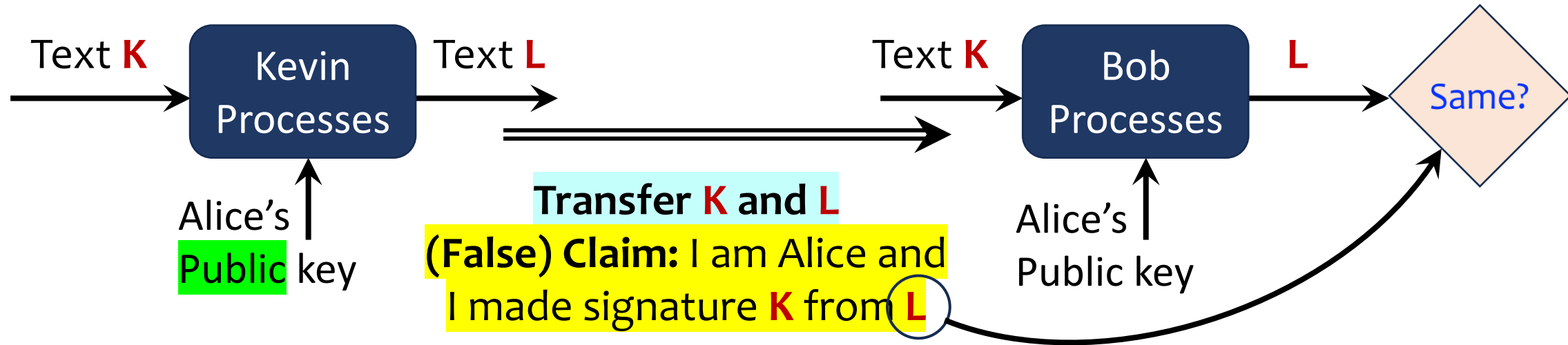


If what Bob decodes (**M**) matches,
then it's legit (integrity) and **Y** is also fine (authenticity)...
authentication and integrity verified?

Thought Question: Bob receives "some" data
Can he now be sure it came from Alice?

Kevin can claim to be Alice

Kevin has no access to Alice's private key – can still cause **trouble!**



Since **L** matches, Bob thinks **L** is a legit message and... sender is Alice

(A little bit of) **Good News:** Kevin can control **K** but not **L**

(Still some) **Bad News:** It's an injection of garbage data

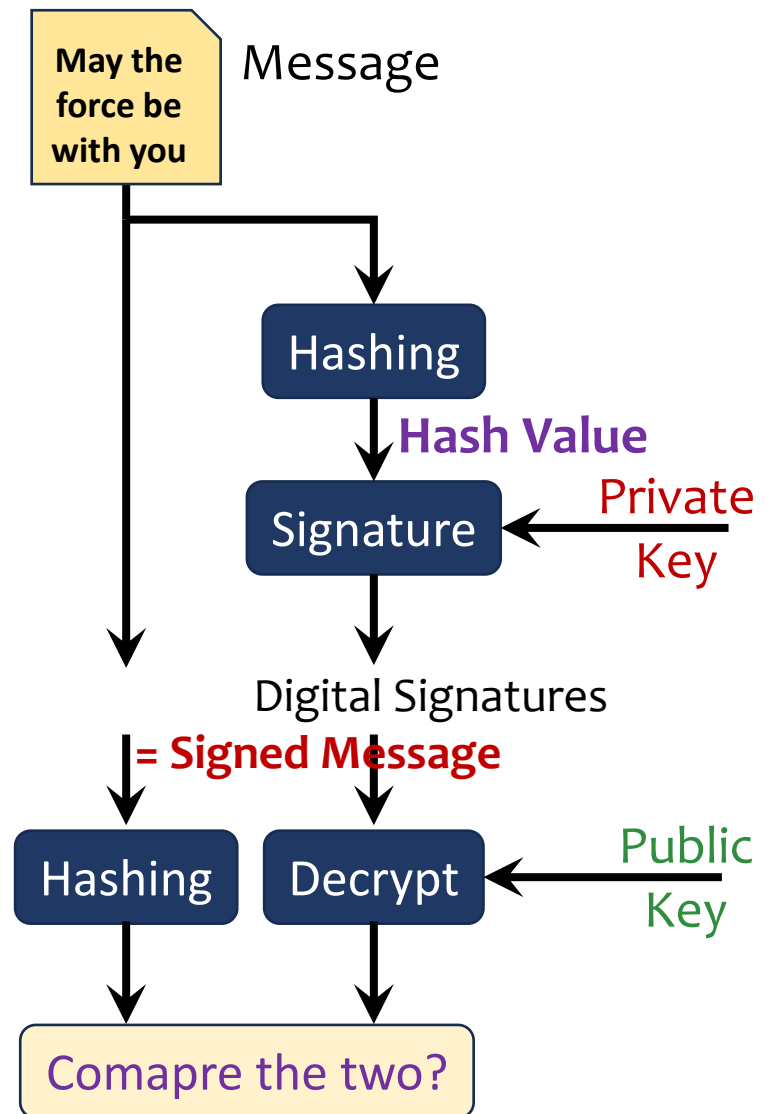
How to avoid this? Try hashing before signing

Public Key Cryptography

- 1) Create a message digest (using cryptographic hashing)
- 2) Encrypt the message digest with your private key
- 3) Send the signatures along with the message

- At the receiver
 - Decrypt the hash value [using public key]
 - Compute message hash value
 - **Compare!**
- Brings about:
 - Sender authentication
 - Message integrity

Kevin (or anyone else) can't insert garbage
Because the hash function is **one-way**

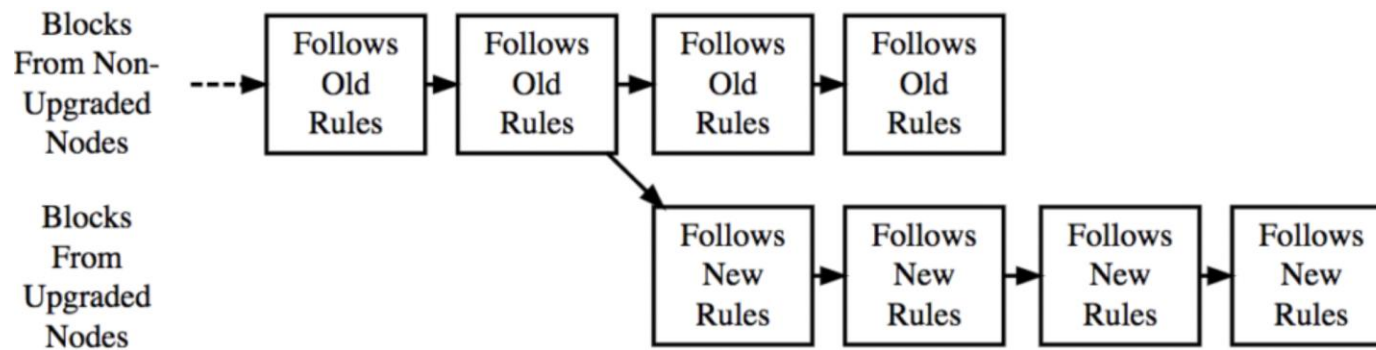


Blockchain Security

- With all the security, authentication, verification, etc., **can we still break a blockchain?**
 - Before answering this, we need to understand/consider a **fork**
- Two miners can almost simultaneously create candidate blocks
 - Broadcasted to all connected nodes – both blocks valid!
 - Added to journals, causing a fork – an **accidental** or **natural** fork
- Over time, one will become part of the ledger (longest chain)
 - Naturally occurring forks are **frequent** and **automatically** resolved
 - When we talk of forks, we don't talk of natural/accidental ones

Forks with a Purpose: Hard Fork

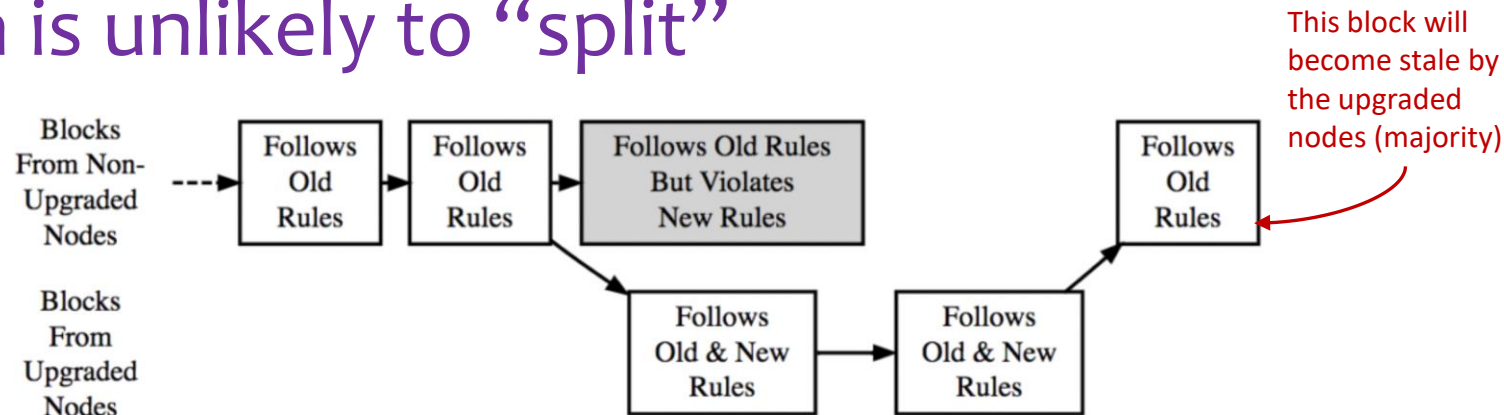
- Loosens the rules for newly created blocks (more liberal rules)
- Ex: increase mining reward, increase the max block size
- A block invalid under old rules may be valid under new rules
- Only nodes that have upgraded will accept the new blocks
- All nodes **MUST** upgrade to avoid a “split” (often, they don’t)



A Hard Fork: Non-Upgraded Nodes Reject The New Rules, Diverging The Chain

Forks with a Purpose: Soft Fork

- Tightens the rules for newly created blocks
- Ex: decreasing mining reward, decrease the max block size
- A block valid under old rules may be invalid under new rules
- All nodes (upgraded or not) will accept the new blocks
- If majority nodes upgraded, old nodes don't need to upgrade
- Blockchain is unlikely to “split”



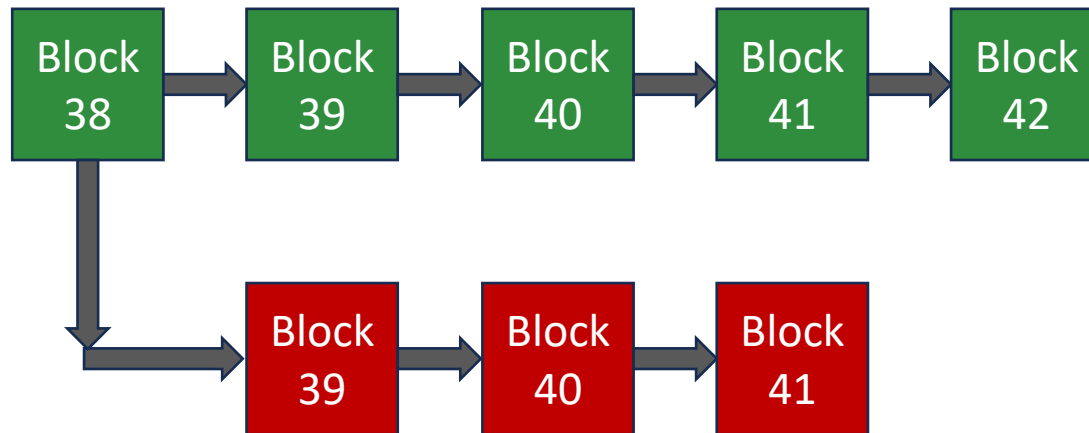
A Soft Fork: Blocks Violating New Rules Are Made Stale By The Upgraded Mining Majority

Lecture 06

Thursday, Feb 5, 2026

The 51% Attack

Truthful miners are adding blocks to the public chain by broadcasting them



Truthful Miners

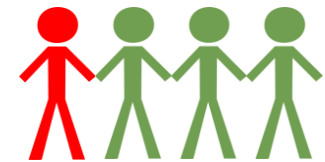
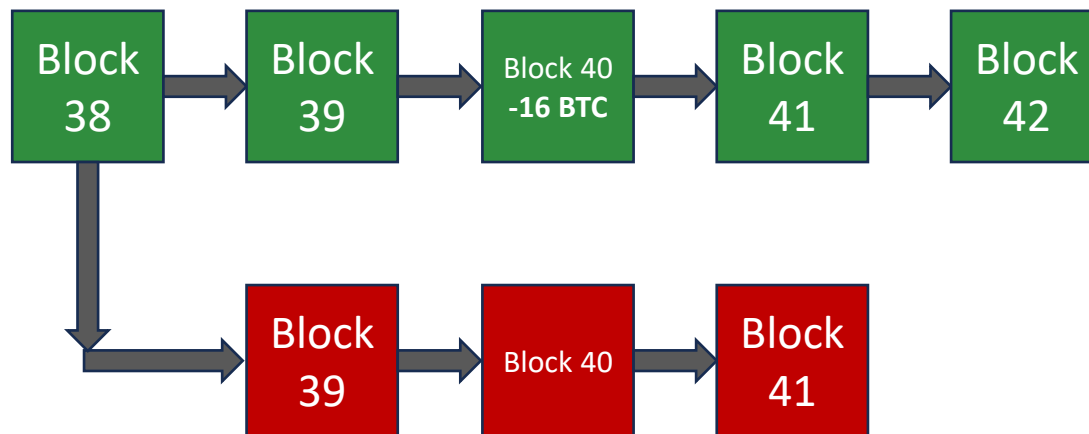


Malicious Miner

Malicious miner: adding blocks to the private chain but **not broadcasting** to the public chain

The 51% Attack

Malicious miner: spends BTC (for buying items) and records on the truthful public chain



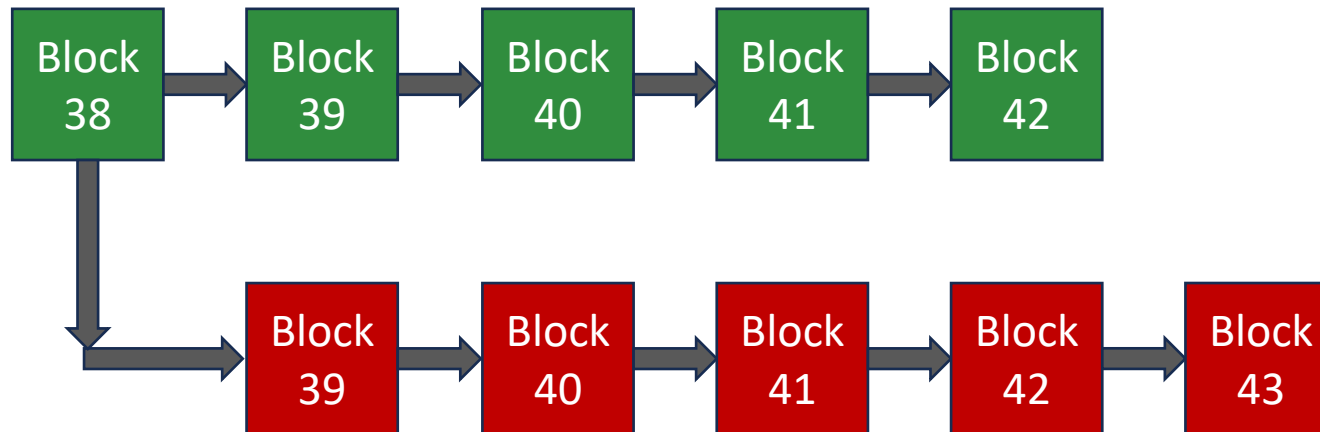
Malicious miner: excludes these transactions from his private chain – still owns BTC there!

The 51% Attack

Truthful miners: adding the blocks to the public truthful chain at their own rate



Hashing Power



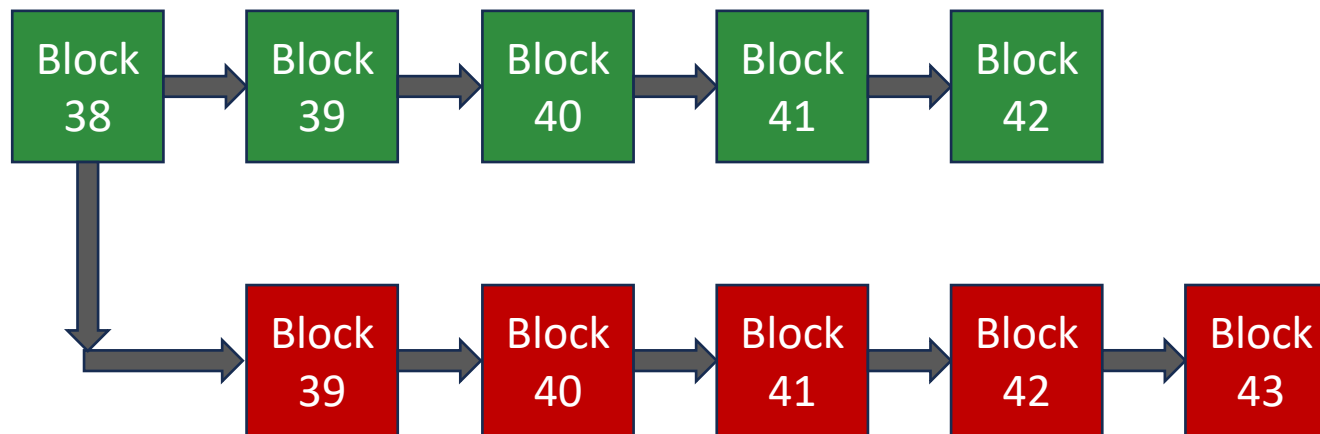
Malicious miner: adding blocks to the private chain **faster** owing to higher compute power



Higher Hashing Power

The 51% Attack

Truthful miners: they must accept the longest chain (following the protocol)



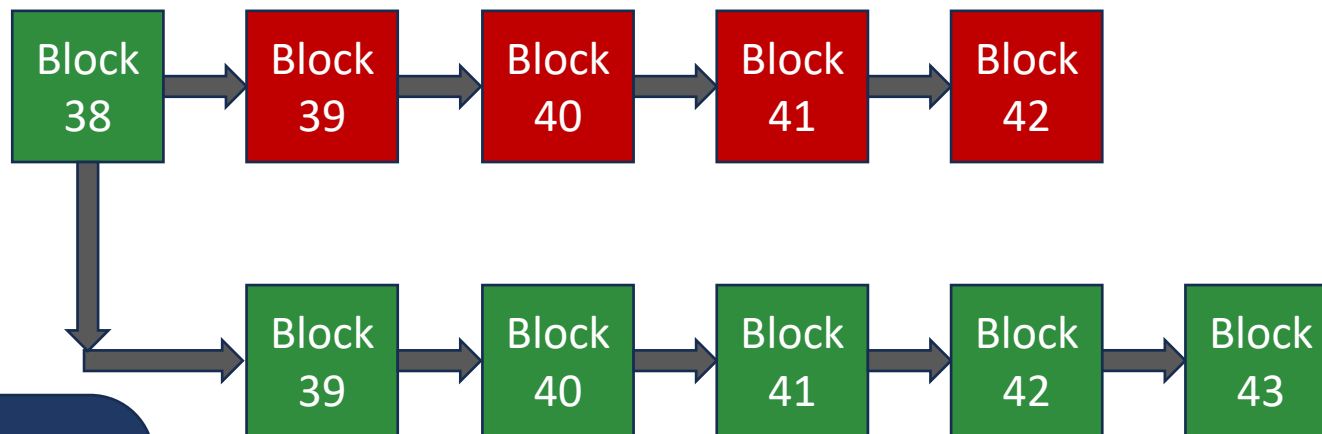
Broadcast

Malicious miner: broadcasts his chain once it is longer than the public one



The 51% Attack

The old public chain is abandoned (including the spending record of the malicious miner)



Everyone agrees on the longest chain



What happens to the transaction in which the malicious miner had spent? Does it go back to the mempool?

Malicious miner: Can “double” spend the BTC that he already spent on the other chain

Evolution of Money

The Evolution of Money

- Before the Money
- Barter System



Barter



- Can Make it work 3-way

The Evolution of Money: Commodity Money

- Divisible
- Stable Value
- **Scarcity is the key**

Precious
Stones



Alcohol



Cocoa Beans



Copper



Gold



Salt



Sea Shells



Silver



Tea



Tobacco



Commodity of Intrinsic Value

- Inherently works peer-to-peer but ...
 - **Cumbersome to use for large purchases**
 - How many silver coins are needed to buy an expensive item?

The Evolution of Money: Representative Money

- Government-produced money
 - Backed by some precious commodity
- Has little to **no intrinsic value** of its own
 - But can be exchanged for a commodity that has some intrinsic value
- Easier to transact, enforced unforgeability

Often backed by Gold or Silver



- Value based on the backing instrument
 - But NOT just “**some common value**” that everyone agrees upon

The US Gold Reserve (Fort Knox, KY)

- 4583 Metric tons (about 56% of US Gold Reserves)
- Federal Reserve Bank building in New York vault
 - Over 6000 Metric tons of Gold
 - More than 95% belongs to other countries



The US Gold Reserve (Fort Knox, KY)

What's inside?



Fort Knox Gold Reserve Value

- Total reserves in USD: \$541B (@ \$118M per Metric ton)
- About 1/8th of NVIDIA's Market Cap (16 Sep 2025)

Last 10 years

Gold Price: slightly < 3x
NVDA stock: slightly > 300x



The evolution of money: Fiat money

- Government-produced money (USD, Euro, PKR)
- Has little to no intrinsic value of its own
 - It may not even be backed by some precious commodity
 - Value is based on what is set by the government
 - Money that the government guarantees as legal tender
- *In 1971, the US severed ties with the Gold Standard*

